

ENONTEKIO, Finland

WMO: 028075

Lat: 68.363N

Lon: 23.424E

Elev: 306

StdP: 97.70

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.8	(c) -29.2	(d) -36.0	(e) 0.1	(f) -32.5	(g) -32.2	(h) 0.2	(i) -29.2	(j) 9.5	(k) -4.1	(l) 8.4	(m) -4.7	(n) 0.5	(o) 210	(p) 0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	24.0	15.7	21.8	14.3	19.2	13.5	16.7	21.7	15.5	19.6	14.4	18.1	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.0	17.7	13.8	10.2	16.7	12.8	9.5	15.7	47.7	21.9	44.2	19.5	41.2	18.1	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.7	-37.1	26.1	3.1	2.3	-39.3	27.7	-41.2	29.1	-42.9	30.4	-45.2	32.1		
			-37.2	18.0	3.1	1.6	-39.4	19.1	-41.2	20.1	-43.0	21.0	-45.2	22.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-13.6	-13.2	-8.8	-2.1	4.3	9.7	13.6	11.2	6.4	-0.9	-7.6	-10.2		
	DBStd	10.91	7.60	7.92	6.43	4.43	4.37	3.75	3.70	3.37	3.53	5.23	6.39	7.25		
	HDD10.0	4205	733	651	582	363	186	49	8	27	115	338	528	626		
	HDD18.3	7014	991	884	841	613	436	259	153	223	357	596	778	884		
	CDD10.0	241	0	0	0	0	8	41	119	64	8	0	0	0		
	CDD18.3	7	0	0	0	0	0	1	6	1	0	0	0	0		
	CDH23.3	81	0	0	0	0	4	4	64	8	0	0	0	0		
	CDH26.7	7	0	0	0	0	0	0	7	1	0	0	0	0		
(14) Wind	WSAvg	2.9	2.4	2.6	3.0	3.0	3.4	3.6	3.1	2.7	2.9	2.7	2.4	2.6		
(15) Precipitation	PrecAvg	510	31	28	26	26	37	60	72	70	43	46	39	33		
	PrecMax	630	61	60	62	66	74	109	132	130	94	96	67	73		
	PrecMin	364	7	6	10	7	8	17	19	36	10	13	11	7		
	PrecStd	79	13	13	13	15	18	28	30	24	22	21	18	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	3.2	5.2	10.9	22.2	24.1	27.8	24.8	17.2	10.2	4.9	4.0		
		MCWB	0.5	1.6	2.1	5.6	13.2	15.2	17.7	17.8	12.3	8.1	4.7	2.2		
	2%	DB	-0.8	0.8	2.9	7.9	18.0	21.1	25.0	21.8	15.0	8.2	3.0	1.8		
		MCWB	-1.1	-0.2	0.7	3.9	10.5	13.6	15.7	14.9	11.1	7.0	2.1	0.9		
	5%	DB	-2.0	-1.0	1.1	5.9	14.8	18.9	23.1	19.1	13.1	7.1	1.1	0.0		
		MCWB	-2.3	-1.5	-0.6	2.7	9.0	12.1	15.0	14.0	10.3	6.3	0.9	-0.3		
	10%	DB	-3.8	-2.9	-0.1	4.2	11.2	16.2	20.8	17.1	11.9	5.8	0.0	-1.0		
		MCWB	-4.1	-3.2	-1.2	1.4	6.8	10.9	14.7	13.1	9.6	5.2	-0.1	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.9	2.4	6.0	13.1	16.8	18.6	18.2	13.2	8.6	4.7	2.4		
		MCDB	0.9	3.0	5.0	10.2	22.3	22.0	23.5	22.7	15.9	9.1	4.8	3.6		
	2%	WB	-1.0	-0.3	0.8	3.9	11.0	14.6	17.1	16.0	12.0	7.2	2.1	0.7		
		MCDB	-0.8	0.3	2.4	6.9	17.0	18.6	22.8	19.6	14.2	8.0	2.5	1.2		
	5%	WB	-2.5	-1.7	-0.3	2.8	9.0	13.2	16.1	14.6	11.0	6.2	0.8	-0.2		
		MCDB	-2.3	-1.4	0.7	5.5	13.6	17.3	21.0	17.7	13.0	6.9	1.1	0.0		
	10%	WB	-4.0	-3.3	-1.4	1.7	7.4	11.8	15.0	13.8	10.1	4.8	-0.1	-1.6		
		MCDB	-3.8	-3.0	-0.4	3.9	10.7	15.7	19.3	16.7	11.7	5.3	0.0	-1.2		
(35) Mean Daily Temperature Range	MDBR		8.2	8.4	10.1	8.6	7.7	7.7	8.5	8.5	7.1	5.6	6.6	7.6		
	5% DB	MCDBR	7.3	7.6	7.7	9.6	13.3	11.5	12.4	11.7	9.0	5.8	5.0	5.7		
		MCWBR	7.0	7.0	6.3	6.5	7.2	5.6	5.3	6.1	5.9	4.9	4.8	5.3		
	5% WB	MCDBR	7.2	7.1	7.0	8.4	11.8	9.1	10.2	9.3	7.3	5.2	4.8	5.4		
		MCWBR	7.0	6.7	6.0	5.9	6.7	5.1	5.1	5.4	5.0	4.5	4.7	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.182	0.240	0.279	0.308	0.328	0.332	0.339	0.325	0.294	0.257	N/A	N/A		
	taud		1.919	2.091	2.188	2.216	2.401	2.490	2.484	2.505	2.561	2.305	N/A	N/A		
	Ebn at Noon		89	549	753	835	865	871	850	824	767	543	N/A	0		
	Edh at Noon		7	45	77	101	95	90	88	77	56	37	N/A	0		
(44) All-Sky Solar Radiation	RadAvg		0.03	0.47	1.98	3.97	5.02	4.94	4.64	3.31	1.78	0.58	0.09	0.00		
	RadStd		0.00	0.04	0.09	0.18	0.49	0.34	0.45	0.23	0.19	0.09	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page